Mathematics Examination 2023

Question 1 (6 marks)

Prove that the sum of the first 1 odd numbers is 1^2.

Show all working.

Question 2 (6 marks)

Prove that the sum of the first 2 odd numbers is 2^2.

Show all working.

Question 3 (6 marks)

Prove that the sum of the first 3 odd numbers is 3^2.

Show all working.

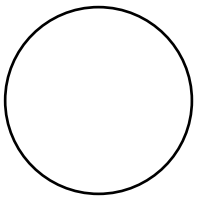
Question 4 (6 marks)

Let f(x) = x^2 + 1 for all real x.

(a) Find f(3).

(b) Show that f(x) is never prime for even x > 2.

Use the diagram below.



(c) Sketch the graph of y = f(x).

Question 5 (6 marks)

Prove that the sum of the first 5 odd numbers is 5^2.

Show all working.